

nonidentical natural frequency case, a similar conclusion is obtained on phase locking. In this case, we also prove that relative phases (synchronization error) can be made arbitrarily small by making the pacemaker strength strong enough. The results are independent of oscillator numbers in the network and are confirmed by numerical simulations.

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Robust State-Dependent Switching of Linear Systems With Dwell Time

Liron I Allerhand and Uri Shaked

Abstract—A state-dependent switching law that obeys a dwell time constraint and guarantees the stability of a switched linear system is designed. Sufficient conditions are obtained for the stability of the switched systems when the switching law is applied in presence of polytopic type parameter uncertainty. A Lyapunov function, in quadratic form, is assigned to each subsystem such that it is non-increasing at the switching instants. During the dwell time, this function varies piecewise linearly in time. After the dwell, the system switches if the switching results in a decrease in the value of the LF. The method proposed is also applicable to robust stabilization via state-feedback. It is further extended to guarantee a bound on the $\mathcal{L}_2$ -gain of the switching system; it is also used in deriving state-feedback control law that robustly achieves a prescribed $\mathcal{L}_2$ -gain bound.

Index Terms—Dwell time, linear switched systems, polytopic uncertainties, state-dependent switching, switching law design.

I. INTRODUCTION

In this work the analysis and synthesis of a family of state-dependent switching laws with dwell time constraints are considered with a focus on stabilization and on achieving a prescribed $\mathcal{L}_2$ -gain bound for switched systems with polytopic type parametric uncertainty. Such systems are encountered, for example, when switching between several different controllers is required [1], and in power systems such as switched converters [2]. The switching of such converters

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is achieved using FET transistors. These transistors have a maximal allowed switching frequency, and thus some dwell time is always encountered in these converters. It has been shown that the transient response of a system can be improved by controller switching [3], [4], and specifically by switching with dwell time [5].

State-dependent switching laws have been first considered in [6]. The method there does not guarantee any minimal dwell time. The practical case, however, is one where some minimal time period between consecutive switchings is required.

Switching with dwell time between given controllers for linear time invariant systems has been considered in [7]–[10], where conditions for stabilization have been obtained. The later results, called Supervisory Control (SC), have been extended to time-varying systems in [11]. In the SC, a given stabilizing controller is assigned to each of the subsystems. The switching between the controllers is done by assigning an observer to each of the subsystems, and the switching is made to the controller that fits the observer with the least error in some sense, under dwell time and hysteresis constraints. It was shown that SC can be used to overcome some of the limitations of traditional adaptive control [12]. Introductions to SC can be found in [1] and [13]. In all these works, SC is dedicated to controller switching based on an estimator bank and it is different from state-dependent switching between subsystems that can also be open-loop unstable.

Switched systems with delay have been studied in [14] and [15]. A Lyapunov-Krasovskii functional is used there to prove the stability and find a bound on the $\mathcal{L}_2$ -gain of the delay switched systems. [14] deals with delay switched systems under an average dwell time constraint and [15] deals with state-dependant switching of delay switched systems.

In the present work, we introduce state-dependent switching under a dwell constraint. An overview of different constraints that may be imposed on the switching law may be found in [16]. We investigate the bound on the achievable $\mathcal{L}_2$ -gain of this system, and find state-feedback controllers that minimize these bounds. The approach we take in the present work stems from the switched systems with dwell time and arbitrary switching approach of [17]. For review of stability of switched systems with dwell time and arbitrary switching see [17], [18], and also [19]. Another approach for stabilizing a class of linear switched systems by synthesis of limit cycles is found in [20]. For a review of the current methods for stabilizing a switched system by designing the switching law, without a dwell time constraint, see the recent book [19] and the references therein. In the later reference the effect of dwell time on the stability of switched system is also investigated.

It has been shown in [21] that the most destabilizing switching law for a switched system with stable subsystems applies, in many cases, a dwell time. Using the same analysis, it is found that the most stabilizing switching law for many switched systems with unstable subsystems obeys some dwell time. Adding a dwell time constraint to a suboptimal switching law may thus achieve better results.

In this work we present a state-dependent switching law that obeys a dwell time constraint with a prescribed dwell time. In practice, a dwell time constraint is mandatory for every switched system, because there is always a limit to the rate in which the physical system can perform switching. The dwell time may be determined according to the capacity of the systems components, or by the designers desire to reduce discontinuities in the systems operation. For example, if the plant is a passenger aircraft the designer may demand a large dwell time constraint to refrain from making the ride bumpy for the passengers.

We first formulate the problem in Section II. In Section III we extend the results of [6] to obtain stabilization results for nominal switched systems with dwell time and in Section IV a method is introduced that can handle polytopic type parameter uncertainty in these systems. Guaranteed $\mathcal{L}_2$ -gain bounds are obtained in Section V. Robust state-

feedback synthesis is introduced in Section VI where both the feedback gains and the switching law are derived that stabilize the uncertain system and minimize the bound on the $\mathcal{L}_2$ -gain of the resulting closed-loop system. Three illustrative examples are given in Section VII.

Notation

Throughout the technical note the superscript $'$ stands for matrix transposition, $\mathcal{R}^n$ denotes the n dimensional Euclidean space and $\mathcal{R}^{n \times m}$ is the set of all $n \times m$ real matrices. For a symmetric $P \in \mathcal{R}^{n \times n}$, $P > 0$ means that it is positive definite. Switching law of a switched system is denoted by $\sigma(t)$. A symmetric matrix $\begin{bmatrix} Q & R \\ R' & P \end{bmatrix}$ is denoted by $\begin{bmatrix} Q & R \\ * & P \end{bmatrix}$. L_i denotes the set of integers $\{1, \dots, i-1, i+1, \dots, M\}$, and $l_1^{(i)}, \dots, l_{M-1}^{(i)}$ denote elements of this set. The scalar τ^- denotes $\lim_{\epsilon \rightarrow 0} (\tau - \epsilon)$ where $\epsilon > 0$.

II. PROBLEM FORMULATION

We consider the following linear switched system:

$$\begin{aligned} \dot{x}(t) &= A_{\sigma(t)}x(t) + B_{\sigma(t)}w(t) + B_{2, \sigma(t)}u(t), \quad x(0) = 0, \\ z(t) &= C_{\sigma(t)}x(t) + D_{\sigma(t)}w(t) + D_{12, \sigma(t)}u(t) \end{aligned} \quad (1)$$

where $x \in \mathcal{R}^n$ is the state, $z \in \mathcal{R}^r$ is the objective vector, $u \in \mathcal{R}^s$ is the control signal and $w \in \mathcal{R}^q$ is an exogenous disturbance in $\mathcal{L}_2$. This system is defined for all $t \geq 0$. The switching rule $\sigma(t) \in \{1 \dots M\}$, for each $t \geq 0$, is such that $\Omega_{\sigma(t)} \in \{\Omega_1, \dots, \Omega_M\}$, where

$$\Omega_{\sigma(t)} = \begin{bmatrix} A_{\sigma(t)} & B_{\sigma(t)} & B_{2, \sigma(t)} \\ C_{\sigma(t)} & D_{\sigma(t)} & D_{12, \sigma(t)} \end{bmatrix}.$$

The polytope Ω_i is defined by

$$\Omega_i = \sum_{j=1}^N \eta^j(t) \Omega_i^{(j)}, \quad \sum_{j=1}^N \eta^j(t) = 1, \quad \eta^j(t) \geq 0 \quad (2)$$

with vertices defined by

$$\Omega_i^{(j)} = \begin{bmatrix} A_i^{(j)} & B_i^{(j)} & B_{2,i}^{(j)} \\ C_i^{(j)} & D_i^{(j)} & D_{12,i}^{(j)} \end{bmatrix}, \quad i = 1, \dots, M \quad (3)$$

where $A_i^{(j)}$, $B_i^{(j)}$, $B_{2,i}^{(j)}$, $C_i^{(j)}$, $D_i^{(j)}$, $D_{12,i}^{(j)}$ are constant matrices of the appropriate dimensions. The above model naturally imposes discontinuity in $\Omega_{\sigma(t)}$ since this matrix jumps instantaneously from Ω_{i_1} to Ω_{i_2} for some $i_1 \neq i_2$ at the switching instances.

Note that this means that each subsystem has a separate polytopic description, which is independent of the description of the other subsystems.

The present work is focused on stabilizing the system via state-feedback control and on achieving a prescribed bound on the $\mathcal{L}_2$ -gain of the system under a dwell time constraint. The latter means that if the dwell time is T , and the switching instants are $\tau_1, \tau_2, \dots$, then $\tau_{h+1} - \tau_h \geq T$, $\forall h \geq 1$.

The measure of performance in this work is thus the bound on the $\mathcal{L}_2$ -gain of the system, from the disturbance w to the objective vector z .

III. NOMINAL TIME SWITCHING STABILITY

Theorem 1: Assume there exist in $\mathcal{R}^{n \times n}$ a collection of positive definite matrices P_i , $i = 1, \dots, M$ and a set of scalars $\lambda_{i,l} \geq 0$, $i = 1 \dots M$, $l \in L_i$, such that

$$P_i A_i + A_i' P_i + \sum_{l \in L_i} \lambda_{i,l} (e^{A_i' T} P_l e^{A_i T} - P_i) < 0, \quad \forall i = 1, \dots, M. \quad (4)$$

Assume also that the nominal system (1) switched at $t = \tau_h$ to $\sigma(\tau_h) = i$. Then, the following law for the next switching at $t = \tau_{h+1}$:

$$\begin{aligned} \sigma(t) &= i \quad \forall t \in [\tau_h, \tau_h + T) \\ \sigma(t) &= i \quad \forall t > \tau_h + T, \\ \text{if } x(t)' P_i x(t) &\leq x(t)' e^{A_i' T} P_l e^{A_l T} x(t), \quad \forall l \in L_i \\ \sigma(\tau_{h+1}) &= \arg \min_l x(\tau_{h+1})' e^{A_l' T} P_l e^{A_l T} x(\tau_{h+1}), \\ &\quad \text{otherwise} \end{aligned} \quad (5)$$

globally asymptotically stabilizes the system (1) with dwell time constraint $T > 0$.

Proof: Let $i_0 = \sigma(0)$, and let $\tau_1, \tau_2, \dots$ be the switching instants, where $\tau_{h+1} - \tau_h \geq T$, $\forall h = 1, 2, \dots$. Choosing the Lyapunov Function (LF) $V(t) = x'(t) P_{\sigma(t)} x(t)$ we have, between $\tau_h + T$ and τ_{h+1} , that

$$\dot{V}(t) = x'(t) (P_i A_i + A_i' P_i) x(t)$$

along the system trajectory $x(t)$, and from the definition of the switching law (5) we have that

$$x'(t) (e^{A_i' T} P_l e^{A_l T} - P_i) x(t) \geq 0, \quad \forall l \in L_i.$$

Hence

$$\dot{V}(t) \leq x'(t) [P_i A_i + A_i' P_i + \sum_{l \in L_i} \lambda_{i,l} (e^{A_l' T} P_l e^{A_l T} - P_i)] x(t)$$

and it follows from (4) that:

$$\begin{aligned} P_i A_i + A_i' P_i + \sum_{l \in L_i} \lambda_{i,l} (e^{A_l' T} P_l e^{A_l T} - P_i) &< 0 \\ \text{and thus } \dot{V}(t) &< 0. \end{aligned}$$

At the switching instant $t = \tau_{h+1}$, $V(\tau_{h+1}^-) = x'(\tau_{h+1}) P_i x(\tau_{h+1})$, and $V(\tau_{h+1} + T) = x'(\tau_{h+1}) e^{A_i' T} P_l e^{A_l T} x(\tau_{h+1})$. Thus, $V(\tau_{h+1} + T) - V(\tau_{h+1}^-) = x'(\tau_{h+1}) (e^{A_i' T} P_l e^{A_l T} - P_i) x(\tau_{h+1})$. From the definition of the switching law we have $x'(\tau_{h+1}) (e^{A_i' T} P_l e^{A_l T} - P_i) x(\tau_{h+1}) < 0$ and therefore $V(\tau_{h+1} + T) - V(\tau_{h+1}^-) < 0$.

We note that the switching strategy of (5) is not claimed to produce an optimal result in any sense. It only provides a possible solution for the stabilization problem. In the case where at least one of the subsystems is stable, stability is achieved by merely switching to this system.

The result of Theorem 1 cannot be easily applied to the uncertain case where the parameters of A_i are only known to reside in Ω_i , $i = 1, \dots, M$. We introduce below a LF based approach to deal with the uncertainty. The results obtained will converge to those of [6] in the limit where the dwell time tends to zero.

IV. ROBUST SWITCHING STABILITY

We first derive the following result, inspired by [22].

Lemma 1: Assume that for some time interval $t \in [t_0, t_f]$, where $\delta = t_f - t_0$ there exist two positive definite matrices P_1 and P_2 of compatible dimensions that satisfy the following:

$$\frac{P_2 - P_1}{\delta} + P_1 A + A' P_1 < 0, \quad \frac{P_2 - P_1}{\delta} + P_2 A + A' P_2 < 0. \quad (6)$$

Then, for the system $\dot{x} = Ax$, the LF $V(t) = x'(t) P(t) x(t)$, with $P(t) = P_1 + (P_2 - P_1)((t - t_0)/\delta)$, is strictly decreasing over the time interval $t \in [t_0, t_f]$.

Proof: The derivative of $V(t)$ is: $\dot{V}(t) = x'(\dot{P} + P A + A' P) x = x'((P_2 - P_1)/\delta + P A + A' P) x$. Since

$$\begin{aligned} \frac{P_2 - P_1}{\delta} + P A + A' P &= \lambda_1 \left[\frac{P_2 - P_1}{\delta} + P_1 A + A' P_1 \right] \\ &\quad + \lambda_2 \left[\frac{P_2 - P_1}{\delta} + P_2 A + A' P_2 \right] \end{aligned}$$

where $\lambda_1 = 1 - (t - t_0)/\delta$ and $\lambda_2 = (t - t_0)/\delta$, we obtain from (6) that $\dot{V}(t) < 0$ in the time interval specified above.

Note that while the latter result has been obtained for a nominal system, the extension to the polytopic uncertainty case is readily achieved by choosing the same P_1 and P_2 for all the vertices of the uncertainty polytope.

Based on Lemma 1 we obtain the following, for a prescribed integer K that is chosen according to the allowed computational complexity. A larger K leads to less conservatism in the conditions.

Theorem 2: Given that the nominal system (1) switched at $t = \tau_h$ to $\sigma(\tau_h) = i$ and assume there exist a collection of positive definite matrices $P_{i,k}$, $i = 1, \dots, M$, $k = 0, \dots, K$ in $\mathcal{R}^{n \times n}$, and a set of scalars $\lambda_{i,l} \geq 0$, $i = 1, \dots, M$, $l \in L_i$, such that

$$\begin{aligned} \frac{P_{i,k+1} - P_{i,k}}{\frac{T}{K}} + P_{i,k} A_i + A_i' P_{i,k} &< 0, \\ \frac{P_{i,k+1} - P_{i,k}}{\frac{T}{K}} + P_{i,k+1} A_i + A_i' P_{i,k+1} &< 0, \\ k &= 0, \dots, K-1 \end{aligned}$$

$$P_{i,K} A_i + A_i' P_{i,K} + \sum_{l \in L_i} \lambda_{i,l} (P_{l,0} - P_{i,K}) < 0. \quad (7)$$

Then, the following law for the next switching at $t = \tau_{h+1}$:

$$\begin{aligned} \sigma(t) &= i \quad \forall t \in [\tau_h, \tau_h + T) \\ \sigma(t) &= i \quad \forall t > \tau_h + T, \\ \text{if } x(t)' P_{i,K} x(t) &\leq x(t)' P_{l,0} x(t), \quad \forall l \in L_i \\ \sigma(\tau_{h+1}) &= \arg \min_l x(\tau_{h+1})' P_{l,0} x(\tau_{h+1}), \quad \text{otherwise} \end{aligned} \quad (8)$$

globally asymptotically stabilizes the system (1) with a constraint of dwell time $T > 0$.

Proof: Let $i_0 = \sigma(0)$, and let $\tau_1, \tau_2, \dots$ be the switching instants, where $\tau_{h+1} - \tau_h \geq T$, $\forall h = 1, 2, \dots$. Define $\tau_{h,k} = \tau_h + kT/K$ for $k \geq 1$, and $\tau_{h,0} = \tau_h$. Note that the dwell time constraint implies $\tau_h + T \leq \tau_{h+1,0} = \tau_{h+1}$. We consider the LF $V(t) = x'(t) P(t) x(t)$, where $P(t)$ is chosen to be

$$P(t) = \begin{cases} P_{i,k} + (P_{i,k+1} - P_{i,k}) \frac{t - \tau_{h,k}}{\frac{T}{K}} & t \in [\tau_{h,k}, \tau_{h,k+1}) \\ P_{i,K} & t \in [\tau_{h,K}, \tau_{h+1,0}) \\ P_{i_0,K} & t \in [0, \tau_1) \end{cases} \quad (9)$$

for $h = 1, 2, \dots$ and where i is the index of the subsystem that is active at time t .

After the dwell time, and before the next switching instant, namely for $t \in [\tau_h + T, \tau_{h+1})$, we have that $V(t) = x(t)' P_{i,K} x(t)$, where $\dot{x}(t) = A_i x(t)$, and where, by the definition of the switching law, $x(t)' P_{i,K} x(t) \leq x(t)' P_{l,0} x(t)$, $\forall l \in L_i$, along the system trajectory $x(t)$. Therefore

$$\begin{aligned} \dot{V}(t) &= x(t)' [P_{i,K} A_i + A_i' P_{i,K}] x(t) \\ &\leq x(t)' [P_{i,K} A_i + A_i' P_{i,K} + \sum_{l \in L_i} \lambda_{i,l} (P_{l,0} - P_{i,K})] x(t). \end{aligned}$$

Before the next switching occurs we have, by (7c), that for any $x(t) \neq 0$ and especially for the trajectory of the system, this expression is

negative. During the dwell time we consider the time intervals $t \in [\tau_{h,k}, \tau_{h,k+1})$ where $\tau_{h,k+1} = \tau_{h,k} + T/K$. The matrix $P(t)$ changes then linearly from $P_{i,k}$ to $P_{i,k+1}$, and, according to Lemma 1, conditions (7a-b) then guarantee that $V(t)$ is decreasing during this time interval. At the switching instant, we have from (8b) that there is at least one subsystem that switching to would decrease the value of the LF. From (8c) we find that the system switches to the subsystem that yields the minimum value for the LF, and therefore the LF is non-increasing at the switching instant.

The method for solving the inequalities in Theorem 2, and in the results that will be derived below, deserves an additional explanation. The main difficulty in the solution of these inequalities is the non-convex nature of these inequalities which prevents application of standard linear matrix inequality solution packages. The fact that $\lambda_{i,l} \geq 0$, $i = 1 \dots M$, $l \in L_i$ are scalars may allow, in the future, efficient computational tools of solution. In the present work we adopt a simpler approach. We restrict ourselves to $\lambda_{i,l} = \bar{\lambda}$, for all $l \in L_i$, $i = 1 \dots M$, and thus remain with only one tuning parameter. We note that using a line search for this tuning parameter, the resulting inequalities can be solved by computationally efficient methods that are available in the literature for solving LMIs.

Lemma 2: For a dwell time $T \rightarrow 0$, the conditions of Theorem 2 recover those of [6].

Proof: We denote $\epsilon_k = k\sqrt{T/K}$, and consider $P_{i,k} = P_{i,0} - \epsilon_k I$, $i = 1 \dots M, k = 0 \dots K$ which implies that $(P_{i,k+1} - P_{i,k})/T/K = -(1/\epsilon_1)I$. Conditions (7a,b) then readily follow. Substituting $P_{i,K} = P_{i,0} - \epsilon_K I$ into (7c) we obtain

$$P_{i,0}A_i + A_i'P_{i,0} + \sum_{l \in L_i} \lambda_{i,l}(P_{l,0} - P_{i,0} + \epsilon_K I) - \epsilon_K(A_i + A_i') < 0$$

which, due to the fact that $\epsilon_K \rightarrow 0$, leads to the condition obtained in [6].

V. $\mathcal{L}_2$ -GAIN

Given the system (1), we seek a sufficient condition for the asymptotic stability of the above system and for the following performance criterion to hold for a prescribed positive scalar γ and the initial condition $x(0) = 0$

$$J = \int_0^\infty (z'z - \gamma^2 w'w)dt \leq 0, \quad \forall w \in \mathcal{L}_2.$$

Denoting

$$\bar{J} = \lim_{t \rightarrow \infty} [V(t) + \int_0^t (z'z - \gamma^2 w'w)ds]$$

we have that $J \leq \bar{J}$, since $V(t) \geq 0 \forall t$. Taking into account that $V(t)$ is differentiable for all t , except for the switching instances, and that $x(0) = 0$, we have

$$\lim_{t \rightarrow \infty} V(t) = \sum_{h=0}^\infty \int_{\tau_h}^{\tau_{h+1}} \dot{V}(t)dt + \sum_{h=1}^\infty (V(\tau_h) - V(\tau_h^-))$$

where $\tau_0 = 0$. From the switching law (8) we find that $V(t)$ is non increasing at the switching instances, and thus $V(\tau_h) - V(\tau_h^-) \leq 0 \forall h > 0$, and

$$\lim_{t \rightarrow \infty} V(t) \leq \sum_{h=0}^\infty \int_{\tau_h}^{\tau_{h+1}} \dot{V}(t)dt. \quad (10)$$

Denoting

$$\begin{aligned} \tilde{J} &= \sum_{h=0}^\infty \int_{\tau_h}^{\tau_{h+1}} \dot{V}(t)dt + \int_0^\infty (z'z - \gamma^2 w'w)ds \\ &= \sum_{h=0}^\infty \int_{\tau_h}^{\tau_{h+1}} (\dot{V}(s) + z'z - \gamma^2 w'w)ds \end{aligned}$$

it is found from (10) that: $J \leq \bar{J} \leq \tilde{J}$. Consequentially, if $\tilde{J} \leq 0$, the $\mathcal{L}_2$ -gain of the system will be less than or equal to γ , since at the switching instances the LF does not increase.

Theorem 3 ($\mathcal{L}_2$ -Gain): Assume that for a prescribed scalar $\gamma > 0$ there exist a collection of positive definite matrices $P_{i,k}$, $i = 1 \dots M$, $k = 0 \dots K$ in $\mathcal{R}^{n \times n}$ and a set of scalars $\lambda_{i,l} \geq 0$, $i = 1 \dots M$, $l \in L_i$, such that for a prescribed scalar $\gamma > 0$

$$\begin{aligned} &\begin{bmatrix} \Delta_{i,k} + P_{i,k}A_i + A_i'P_{i,k} & P_{i,k}B_i & C_i' \\ * & -\gamma^2 I & D_i' \\ * & * & -I \end{bmatrix} < 0, \\ &\begin{bmatrix} \Delta_{i,k} + P_{i,k+1}A_i + A_i'P_{i,k+1} & P_{i,k+1}B_i & C_i' \\ * & -\gamma^2 I & D_i' \\ * & * & -I \end{bmatrix} < 0, \\ &\begin{bmatrix} P_{i,K}A_i + A_i'P_{i,K} + \bar{\Gamma}_i & P_{i,K}B_i & C_i' \\ * & -\gamma^2 I & D_i' \\ * & * & -I \end{bmatrix} < 0 \end{aligned} \quad (11)$$

where $k = 0 \dots K-1$, $\bar{\Gamma}_i = \sum_{l \in L_i} \lambda_{i,l}(P_{l,0} - P_{i,K})$ and $\Delta_{i,k} = (P_{i,k+1} - P_{i,k})/T/K$.

Then, the switching law (8) globally asymptotically stabilizes the system (1), with a constraint of dwell time $T > 0$, and it guarantees an $\mathcal{L}_2$ -gain bound of γ .

Proof: Let $P(t)$ be as in (9). Condition (11a) guarantees that the LF is positive for any $x \neq 0$. From the standard derivation of the BRL for linear systems [23], we have that the following conditions are equivalent:

$$\dot{V} + z'z - \gamma^2 w'w < 0,$$

$$\begin{bmatrix} \dot{P}(t) + P(t)A + A'P(t) & P(t)B & C' \\ * & -\gamma^2 I & D^T \\ * & * & -I \end{bmatrix} < 0 \quad (12)$$

where the matrices A, B, C, D are as in (1). Following the same method that has been used in the proof of Theorem 1, conditions (11a-b) guarantee that (12b) is satisfied during the dwell time, and condition (11c) guarantees that (12b) holds after the dwell time. The switching law (8) guarantees that the LF is non-increasing at the switching instants. We therefore have that $\tilde{J}$ is negative and thus the $\mathcal{L}_2$ -gain of the system (1) is less than γ .

Note that in Theorem 3, γ can be taken as a decision variable and it is possible to minimize it by solving the LMIs. Therefore, it is possible to minimize the bound on the $\mathcal{L}_2$ -gain using our method. This applies to all the following results in this work.

The conditions of Theorem 3 are unsuitable for design by state-feedback since they lead to non-convex conditions. A dual result that is applicable to this design is derived next.

Corollary 1 (Dual $\mathcal{L}_2$ -Gain): Assume that for a prescribed positive scalar γ there exist a collection of positive definite matrices $Q_{i,k}$, $i =$

$1, \dots, M$, $k = 0, \dots, K$ in $\mathcal{R}^{n \times n}$, where K is a prechosen integer, and a set of scalars $\lambda_{i,l} \geq 0$, $i = 1 \dots M$, $l \in L_i$, such that

$$\begin{bmatrix} \Delta_{i,k}^d + A_i Q_{i,k} + Q_{i,k} A_i' & * & * \\ B_i' & -\gamma^2 I & * \\ C_i Q_{i,k} & D_i & -I \end{bmatrix} < 0, \\ \begin{bmatrix} \Delta_{i,k}^d + A_i Q_{i,k+1} + Q_{i,k+1} A_i' & * & * \\ B_i' & -\gamma^2 I & * \\ C_i Q_{i,k+1} & D_i & -I \end{bmatrix} < 0, \quad k = 0, \dots, K-1 \\ \begin{bmatrix} A_i Q_{i,K} + Q_{i,K} A_i' + \Gamma_i & * & * & * & * \\ B_i' & -\gamma^2 I & * & * & * \\ C_i Q_{i,K} & D_i & -I & * & * \\ Q_{i,K} & 0 & 0 & \frac{-1}{\lambda_{i,l_1^{(i)}}} Q_{l_1^{(i)},0} & * \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ Q_{i,K} & \dots & \dots & \frac{-1}{\lambda_{i,l_{M-1}^{(i)}}} Q_{l_{M-1}^{(i)},0} & * \end{bmatrix} < 0 \quad \forall l = 1, \dots, i-1, i+1, \dots, M \quad (13)$$

where: $\Gamma_i = -Q_{i,K} \sum_{l \in L_i} \lambda_{i,l}$ and $\Delta_{i,k}^d = -(Q_{i,k+1} - Q_{i,k})/T/K$.

Then, the switching law (8), where we substitute $P_{i,K} = Q_{i,K}^{-1}$ and $P_{l,0} = Q_{l,0}^{-1}$, globally asymptotically stabilizes the system (1), with the constraint of dwell time $T > 0$, and it guarantees an $\mathcal{L}_2$ -gain bound of γ for this system.

Proof: Let $Q(t) = P^{-1}(t)$. We depart from the linearity in time assumption on $P(t)$ and assign this linearity to $Q(t)$, namely

$$Q(t) = \begin{cases} Q_{i,k} + (Q_{i,k+1} - Q_{i,k}) \frac{t - \tau_{h,k}}{T/K} & t \in [\tau_{h,k}, \tau_{h,k+1}) \\ Q_{i,K} & t \in [\tau_{h,K}, \tau_{h+1,0}) \\ Q_{i_0,K} & t \in [0, \tau_1) \end{cases}$$

where $h = 1, 2, \dots$. We also consider next the dual condition of (12b). From [23] we have that the following conditions are equivalent:

$$\begin{aligned} \dot{V} + z'z - \gamma^2 w'w &< 0, \\ \begin{bmatrix} \dot{P}(t) + P(t)A + A'P(t) & P(t)B & C' \\ * & -\gamma^2 I & D' \\ * & * & -I \end{bmatrix} &< 0, \\ \begin{bmatrix} -\dot{Q}(t) + AQ(t) + Q(t)A' & B & C'Q(t) \\ * & -\gamma^2 I & D' \\ * & * & -I \end{bmatrix} &< 0 \end{aligned}$$

where $Q(t) = P(t)^{-1}$.

Since $Q(t)$ is now piecewise linear in time, we obtain conditions (13a,b), by applying the same method that has been used in the proof of Theorem 3.

In order to derive (13c), we multiply (11c) by $\begin{bmatrix} Q_{i,K} & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}$ from both sides. Since after the dwell time both P and Q are constant, until the next switching occurs, the latter multiplication leads to the following inequality:

$$\begin{bmatrix} A_i Q_{i,K} + Q_{i,K} A_i' + \Xi & B_i & Q_{i,K} C_i' \\ * & -\gamma^2 I & D_i' \\ * & * & -I \end{bmatrix} < 0$$

$$\begin{aligned} \text{where } \Xi &= Q_{i,K} [\sum_{l \in L_i} \lambda_{i,l} (Q_{l,0}^{-1} - Q_{i,K}^{-1})] Q_{i,K} \\ &= Q_{i,K} [\sum_{l \in L_i} \lambda_{i,l} Q_{l,0}^{-1}] Q_{i,K} - Q_{i,K} \sum_{l \in L_i} \lambda_{i,l}. \end{aligned}$$

By applying Schurs complements to the first term in the right side of the latter equality, (13c) is obtained.

Remark 1: The above conditions are all affine in the systems matrices, and they can, therefore, be readily applied to the uncertain case simply by solving these conditions for all the vertices of the uncertainty polytope, namely by replacing A_i , B_i , $B_{2,i}$, C_i , D_i , $D_{12,i}$ with $A_i^{(j)}$, $B_i^{(j)}$, $B_{2,i}^{(j)}$, $C_i^{(j)}$, $D_i^{(j)}$, $D_{12,i}^{(j)}$, respectively, and by solving for all $j = 1 \dots N$.

Remark 2: Corollary 1 provides also an alternative sufficient condition for stability by taking $\gamma \rightarrow \infty$. The application of this condition in the uncertain case may lead to results that differ from those obtained by Theorem 2.

VI. STATE-FEEDBACK

The synthesis of state-feedback controllers is achieved by replacing $A_i^{(j)}$ and $C_i^{(j)}$ by $A_i^{(j)} + B_{2,i}^{(j)} G_{i,k}$ and $C_i^{(j)} + D_{12,i}^{(j)} G_{i,k}$, respectively. These controllers are time-varying even between the switching instances. This demand for time-varying controllers will be removed in the sequel. By applying the latter substitution to (13), and denoting $Y_{i,k} = G_{i,k} Q_{i,k}$, the following result is obtained for a prechosen integer K .

Corollary 2 (Quadratic State-Feedback): Assume that for a prescribed scalar $\gamma > 0$ there exist: a collection of matrices $Q_{i,k} > 0 \in \mathcal{R}^{n \times n}$, and $Y_{i,k} \in \mathcal{R}^{s \times n}$, $i = 1, \dots, M$, $k = 0, \dots, K$, and a set of scalars $\lambda_{i,l} \geq 0$, $i = 1 \dots M$, $l \in L_i$, such that

$$\begin{aligned} \begin{bmatrix} \Delta_{i,k} + \Phi_{i,h} & * & * \\ B_{2,i}^{(j)'} & -\gamma^2 I & * \\ C_i^{(j)} Q_{i,k} + D_{12,i}^{(j)} Y_{i,k} & D_i^{(j)} & -I \end{bmatrix} &< 0, \quad k = 0, \dots, K-1, \\ h &= k, k+1, \\ \begin{bmatrix} \Phi_{i,K} + \Gamma_i & * & * & * & * \\ B_{2,i}^{(j)'} & -\gamma^2 I & * & * & * \\ C_i^{(j)} Q_{i,K} + D_{12,i}^{(j)} Y_{i,K} & D_i^{(j)} & -I & * & * \\ Q_{i,K} & 0 & 0 & \frac{-1}{\lambda_{i,l_1^{(i)}}} Q_{l_1^{(i)},0} & * \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ Q_{i,K} & \dots & \dots & \frac{-1}{\lambda_{i,l_{M-1}^{(i)}}} Q_{l_{M-1}^{(i)},0} & * \end{bmatrix} &< 0 \end{aligned} \quad (14)$$

where

$$\begin{aligned} \Phi_{i,k} &= A_i^{(j)} Q_{i,k} + Q_{i,k} A_i^{(j)'} + B_{2,i}^{(j)} Y_{i,k} + Y_{i,k}' B_{2,i}^{(j)'} \\ \Gamma_i &= -Q_{i,K} \sum_{l \in L_i} \lambda_{i,l}, \text{ and } \Delta_{i,k} = -\frac{Q_{i,k+1} - Q_{i,k}}{T/K}. \end{aligned}$$

Then, the switching law (8), where we substitute $P_{i,K} = Q_{i,K}^{-1}$ and $P_{l,0} = Q_{l,0}^{-1}$, and the feedback law $u(t) = G(t)x(t)$, where $G(t)$ is given by

$$G(t) = \begin{cases} \left[Y_{i,k} + \frac{t - \tau_{h,k}}{T/K} (Y_{i,k+1} - Y_{i,k}) \right] \left[Q_{i,k} + \frac{t - \tau_{h,k}}{T/K} (Q_{i,k+1} - Q_{i,k}) \right]^{-1}, & t \in [\tau_{h,k}, \tau_{h,k+1}) \\ Y_{i,K} Q_{i,K}^{-1}, & t \in [\tau_{h,K}, \tau_{h+1,0}) \\ Y_{i_0,K} Q_{i_0,K}^{-1}, & t \in [0, \tau_1) \end{cases} \quad (15)$$

globally asymptotically stabilize the system (1) with the uncertainty (2), (3) and with the constraint of dwell time $T > 0$. The latter control law also guarantees an $\mathcal{L}_2$ -gain bound of γ for the switching system.

Proof: In order to preserve the convexity for time values in the interval $[\tau_{h,k}, \tau_{h,k+1})$, we choose: $Y(t) = Y_{i,k} + (t - \tau_{h,k})/(T/K)(Y_{i,k+1} - Y_{i,k})$. The result then follows from the fact that $G_i(t) = Y_i(t)Q_i^{-1}(t)$.

The state-feedback law of Corollary 2 may not be easy to implement. In order to obtain a simpler feedback gain, we introduce an alternative method to guarantee the $\mathcal{L}_2$ -gain of the system, recalling that $Q(t)$ must be time-varying. This method will allow us to obtain simpler controllers, by applying the method of [24].

Corollary 3 ($\mathcal{L}_2$ -Gain): Assume that for prescribed scalar $\gamma > 0$ and integer K there exist a collection of matrices $Q_{i,k} > 0$, $Z_{i,k}$, $H_{i,k} \in \mathcal{R}^{n \times n}$, $i = 1, \dots, M, k = 0, \dots, K$, and a set of scalars $\lambda_{i,l} \geq 0$, $i = 1 \dots M$, $l \in L_i$, such that

$$\begin{bmatrix} \Delta_{i,k}^{(j)} + A_{i,k}^{(j)} Z_{i,h} + Z_{i,h}' A_{i,k}^{(j)'} & * & * & * \\ B_{i,k}^{(j)'} & -\gamma^2 I & * & * \\ C_{i,k}^{(j)} Z_{i,h} & D_{i,k}^{(j)} & -I & * \\ Z_{i,h} - Q_{i,h} - H_{i,h}' A_{i,k}^{(j)'} & 0 & -H_{i,h}' C_{i,k}^{(j)'} & -\hat{H}_{i,h} \end{bmatrix} < 0, \quad k = 0, \dots, K-1, \quad h = k, k+1, \quad (16)$$

$$\begin{bmatrix} \Psi_{i,j} & * & * & * & * \\ B_{i,j}^{(j)'} & -\gamma^2 I & * & * & * \\ C_{i,j}^{(j)} Z_{i,K} & D_{i,j}^{(j)} & -I & * & * \\ \psi_{i,j} & 0 & -H_{i,K}' C_{i,j}^{(j)'} & -\hat{H}_{i,K} & * \\ Q_{i,K} & 0 & 0 & \frac{-1}{\lambda_{i,l(i)}^{(i)}} Q_{l(i),0}^{(i)} & * \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ Q_{i,K} & & & \frac{-1}{\lambda_{i,l(i)}^{(i)}} Q_{l(i),M-1}^{(i)} & 0 \end{bmatrix} < 0 \quad (16)$$

where

$$\Gamma_i = -Q_{i,K} \sum_{l \in L_i} \lambda_{i,l}, \quad \Delta_{i,k}^{(j)} = -\frac{Q_{i,k+1} - Q_{i,k}}{\frac{T}{K}},$$

$$\hat{H}_{i,h} = H_{i,h} + H_{i,h}', \quad \Psi_{i,j} = A_{i,K}^{(j)} Z_{i,K} + Z_{i,K}' A_{i,K}^{(j)'} + \Gamma_i, \\ \psi_{i,j} = Z_{i,K} - Q_{i,K} - H_{i,K}' A_{i,K}^{(j)'}$$

Then, the switching law (8), where we substitute $P_{i,K} = Q_{i,K}^{-1}$ and $P_{l,0} = Q_{l,0}^{-1}$, globally asymptotically stabilizes the system (1) with the uncertainty (2), (3), and with a constraint of dwell time $T > 0$. It also guarantees a bound γ for the $\mathcal{L}_2$ -gain of the system.

Proof: We denote

$$R = \begin{bmatrix} I & 0 & 0 & -A_{i,K}^{(j)} \\ 0 & I & 0 & 0 \\ 0 & 0 & I & -C_{i,K}^{(j)} \\ 0 & 0 & 0 & I \end{bmatrix} \quad \text{and} \quad \bar{R} = \begin{bmatrix} I & 0 & 0 & -A_{i,K}^{(j)} & 0 & \dots \\ 0 & I & 0 & 0 & 0 & \dots \\ 0 & 0 & I & -C_{i,K}^{(j)} & 0 & \dots \\ 0 & 0 & 0 & I & 0 & \dots \\ 0 & 0 & 0 & 0 & I & \ddots \\ \vdots & \vdots & \vdots & \ddots & \ddots & \ddots \end{bmatrix}.$$

The matrix $\bar{R}$ is a square matrix with $(M+1)m + r + q$ rows. Multiplying (16a) by $\bar{R}$, from the left, and by $\bar{R}'$, from the right, and multiplying (16b) by $\bar{R}$, from the left, and by $\bar{R}'$, from the right, the conditions of Corollary 1 are readily obtained.

We note that the method of [24] is usually used to reduce the conservatism entailed in applying a single quadratic LF by using a parameter dependent LF. In our case, since the switching law depends on the LF, one cannot apply a parameter dependent LF without a perfect knowledge of the parameters. Still, the introduction of the slack variables $Z_{i,h}$, $H_{i,h}$ allows one to obtain simpler controllers (e.g., constant feedback gains between switching).

By replacing $A_i^{(j)}$ and $C_i^{(j)}$ by $A_i^{(j)} + B_{2,i}^{(j)} G_{i,k}$ and $C_i^{(j)} + D_{12,i}^{(j)} G_{i,k}$, respectively, denoting $Y_{i,k} = G_{i,k} Z_{i,k}$, and demanding $H_{i,k} = \eta Z_{i,k}$ the following result is obtained for a prechosen integer K and prescribed $\gamma > 0$.

Corollary 4 (State-Feedback): Assume there exist: a collection of matrices $Q_{i,k} > 0$, $Z_{i,k}$, $H_{i,k} \in \mathcal{R}^{n \times n}$, $Y_{i,k} \in \mathcal{R}^{s \times n}$, $i = 1, \dots, M$, $k = 0, \dots, K$, and a set of scalars $\lambda_{i,l} \geq 0$, $i = 1 \dots M$, $l \in L_i$, such that, for a tuning scalar parameter η , the following holds:

$$\begin{bmatrix} \Delta_{i,k,j} + \Phi_{i,h,j} & * & * & * \\ B_{i,j}^{(j)'} & -\gamma^2 I & * & * \\ \Upsilon_{i,h,j} & D_{i,j}^{(j)} & -I & * \\ \psi_{i,h,j} & 0 & \bar{\Upsilon}_{i,h,j} & -\hat{H}_{i,h} \end{bmatrix} < 0, \quad k = 0, \dots, K-1, \quad h = k, k+1, \quad (17)$$

$$\begin{bmatrix} \Psi_{i,j} & * & * & * & * \\ B_{i,j}^{(j)'} & -\gamma^2 I & * & * & * \\ \Upsilon_{i,K} & D_{i,j} & -I & * & * \\ \psi_{i,K,j} & 0 & \bar{\Upsilon}_{i,K} & -\hat{H}_{i,K} & * \\ Q_{i,K} & 0 & 0 & 0 & \frac{-1}{\lambda_{i,l(i)}^{(i)}} Q_{l(i),0}^{(i)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ Q_{i,K} & & & \frac{-1}{\lambda_{i,l(i)}^{(i)}} Q_{l(i),M-1}^{(i)} & 0 \end{bmatrix} < 0 \quad (17)$$

where

$$H_{i,k} = \eta Z_{i,k}, \\ \Phi_{i,h,j} = A_{i,K}^{(j)} Z_{i,h} + Z_{i,h}' A_{i,K}^{(j)'} + B_{2,i}^{(j)} Y_{i,h} + Y_{i,h}' B_{2,i}^{(j)'}, \\ \Gamma_i = -Q_{i,K} \sum_{l \in L_i} \lambda_{i,l}, \quad \Delta_{i,k,j} = -\frac{Q_{i,k+1} - Q_{i,k}}{\frac{T}{K}}, \\ \Psi_{i,j} = A_{i,K}^{(j)} Z_{i,K} + Z_{i,K}' A_{i,K}^{(j)'} + B_{2,i}^{(j)} Y_{i,K} + Y_{i,K}' B_{2,i}^{(j)'} + \Gamma_i, \\ \Upsilon_{i,h,j} = C_{i,K}^{(j)} Z_{i,h} + D_{12,i}^{(j)} Y_{i,h}, \\ \psi_{i,h,j} = Z_{i,h} - Q_{i,h} - H_{i,h}' A_{i,K}^{(j)'} - \eta Y_{i,h}' B_{2,i}^{(j)'}, \\ \bar{\Upsilon}_{i,h,j} = -H_{i,h}' C_{i,K}^{(j)'} - \eta Y_{i,h}' D_{12,i}^{(j)'}, \quad \hat{H}_{i,h} = H_{i,h} + H_{i,h}'.$$

Then, the switching law (8), where we substitute $P_{i,K} = Q_{i,K}^{-1}$ and $P_{l,0} = Q_{l,0}^{-1}$, and the feedback law $u(t) = G(t)x(t)$, where $G(t)$ is given as in (15) with $Z_{i,k}$ replacing $Q_{i,k}$, globally asymptotically stabilize the system (1) with the uncertainty (2), (3), and with a constraint of dwell time $T > 0$. They also guarantees a bound γ for the $\mathcal{L}_2$ -gain of the system.

The matrices $Q_i(t)$ must be time-varying; one can require, however, $Z_i(t)$ and $Y_i(t)$ to be constant between the different switching instances, namely $Z_{i,k_1} = Z_{i,k_2}$, $Y_{i,k_1} = Y_{i,k_2} \quad \forall k_1, k_2 = 0 \dots K$, thus obtaining a controller that is constant between the switching instances. Alternatively, one can obtain a controller that changes piecewise linearly by requiring only $Z_i(t)$ to be constant between the different switching instances, namely $Z_{i,k_1} = Z_{i,k_2}$, $\forall k_1, k_2 = 0 \dots K$.

VII. EXAMPLES

Example 1 ($\mathcal{L}_2$ -Gain): We consider the nominal system of [6] where

$$A_1 = \begin{bmatrix} 0 & 1 \\ 2 & -9 \end{bmatrix}, \quad \text{and} \quad A_2 = \begin{bmatrix} 0 & 1 \\ -2 & 8 \end{bmatrix}.$$

These subsystems are both unstable. Using Theorem 2, this system is found to be stabilized by the switching law (8), with a dwell time of 0.002 s. and $K = 1$, taking $\lambda_{1,2} = \lambda_{2,1} = 300$.

Considering the nominal system (1) with A_1 and A_2 as before and

$$B_1 = B_2 = \begin{bmatrix} 0 \\ 0.1 \end{bmatrix}, \quad C_1 = C_2 = [0.1 \quad 0], \quad D_1 = D_2 = 0.$$

The system is found by Theorem 3 to be stabilized by the switching law (8), with a dwell time of 0.001 s. and $K = 1$ with an $\mathcal{L}_2$ -gain bound of 2.65, using $\lambda_{1,2} = \lambda_{2,1} = 360$. Similarly, taking $K = 5$ we obtain $\gamma = 1.94$, and for $K = 10$ we obtain $\gamma = 1.87$.

Example 2 (State-Feedback): We consider the system (1) with

$$A_1 = \begin{bmatrix} 0 & 1 \\ 2 + \delta & -9 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 1 \\ -2 + \delta & 8 \end{bmatrix},$$

$$B_1 = B_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad B_{2,1} = B_{2,2} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad C_1 = C_2 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},$$

$$D_1 = D_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \text{and} \quad D_{12,1} = D_{12,2} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

where $|\delta| \leq 50$ is an uncertain scalar parameter. Seeking state-feedback gains that are constant between the switching instances, this system is found, using Corollary 4, to be stabilized by the switching law (8), for a dwell time of 0.001 s. and $K = 1$, with an $\mathcal{L}_2$ -gain bound of 0.13, by taking $\lambda_{1,2} = \lambda_{2,1} = 5$ and $\eta = 0.005$. The corresponding feedback gains are $G_1 = [-21.9226 \quad -0.7509]$ and $G_2 = [-27.1678 \quad -20.5190]$. We note that allowing the feedback gains to be time-varying during the dwell time does not improve the minimal bound on γ . The $\mathcal{L}_2$ -gain bound that can be guaranteed by a time-varying controller using Corollary 2 is 0.35.

Example 3 ($\mathcal{L}_2$ -Gain): We consider a Boost converter which is a switched power converter whose purpose is to amplify DC voltage with high efficiency [2]. Switching of such converters is achieved by using FET transistors. In FET transistors there is a limit to the allowed switching frequency, a fact which imposes a minimum required dwell time constraint on these converters. The state vector of the system is $x = [I_L \ V_C \ 1]'$, where I_L is the inductors current, V_C is the capacitors voltage, and the third state is approximately one during the systems activity. The objective vector is the output voltage, measured on the load. The disturbance in this description is an output current disturbance. The first subsystem describes the system dynamics when the inductor is connected to the load, and the second is when the inductor is connected to the ground. We have [2]

$$A_1 = \begin{bmatrix} -\frac{r_L}{L} & -\frac{1}{L} & \frac{V_{in}}{L} \\ \frac{1}{C} & -\frac{1}{C} \frac{1}{r_o} & 0 \\ 0 & 0 & -\epsilon \end{bmatrix}, \quad A_2 = \begin{bmatrix} -\frac{r_L}{L} & 0 & \frac{V_{in}}{L} \\ 0 & -\frac{1}{C} \frac{1}{r_o} & 0 \\ 0 & 0 & -\epsilon \end{bmatrix},$$

$$B_1 = B_2 = \begin{bmatrix} 0 \\ \frac{1}{C} \\ 0 \end{bmatrix}, \quad C_1 = C_2 = [0 \quad 1 \quad -V_{des}],$$

$$D_1 = D_2 = 0$$

with $V_{in} = 100$ V, $r_L = 2 \ \Omega$, $L = 10^{-4}$ H, $r_o \in [50 - 100] \ \Omega$, $C = 4.7 \cdot 10^{-4}$ F, $V_{des} = 200$ V, $\epsilon = 10^{-4}$. In the later, L is the inductor inductance, r_L is the inductor serial resistance, C is the capacitor capacitance, V_{in} is the input voltage, V_{des} is the desired output voltage, and r_o is the loads resistance.

Using Theorem 3 this system is shown to achieve an $\mathcal{L}_2$ -gain bound of 5.41 by the switching law (8), for a given dwell time of $5 \cdot 10^{-4}$ s. This is found for $K = 1$, with $\lambda_{1,2} = \lambda_{2,1} = 2680$. We note that taking a smaller dwell time or eliminating it altogether leads to a smaller bound on the $\mathcal{L}_2$ -gain.

VIII. CONCLUSION

A new method for the analysis and synthesis of state-dependent switching law under a dwell time constraint is introduced, using a switching dependent LF of quadratic structure. This method enables the treatment of systems with polytopic type parameter uncertainties, and it can also be easily extended to treat norm-bounded parameter uncertainties. When the dwell time tends to zero, our results converge to those of [6]. The new method is extended to verify bounds on the $\mathcal{L}_2$ -gain of the switching system and to derive a state-feedback control law that robustly stabilizes the system and guarantees a prescribed $\mathcal{L}_2$ -gain bound for the closed-loop system.

Corollary 1 is introduced in order to provide a simple platform for solving the state-feedback control problem. Application of its inequalities to the derivation of the bound on the $\mathcal{L}_2$ -gain of the system may lead to results that are different from those obtained by Theorem 2. In the latter derivation, the results of Corollary 1 may be quite conservative when dealing with unstable switched systems.

Deliberately introducing a dwell time constraint in the switching law can, in some cases, lead to a reduced bound on the $\mathcal{L}_2$ -gain of the system, e.g., Example 3. This is in line with the result of [21], and it is probably due to the fact that the switching law is not optimal.

In the example section, it has been demonstrated that, in some cases, addition of a dwell time constraint improves the systems performance. This may happen because the switching law that is presented in this work is suboptimal, and thus adding or changing the dwell time may in some cases lead to a smaller $\mathcal{L}_2$ -gain.

The results in this work have been focused on switching between subsystems. Their extension to switching controllers is a subject for a further research.

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Virtual Holonomic Constraints for Euler–Lagrange Systems

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Abstract—This technical brief investigates *virtual holonomic constraints* for Euler–Lagrange systems with n degrees-of-freedom and $n - 1$ controls. In our framework, a virtual holonomic constraint is a relation specifying $n - 1$ configuration variables in terms of a single angular configuration variable. The enforcement by feedback of such a constraint induces a desired repetitive behavior in the system. We give conditions under which a virtual holonomic constraint is feasible, i.e., it can be made invariant by feedback, and it is stabilizable. We provide sufficient conditions under which the dynamics on the constraint manifold correspond to an Euler–Lagrange system. These ideas are applied to the problem of swinging up an underactuated pendulum while guaranteeing that the second link does not fall over.

Index Terms—Mechanical systems, nonlinear control systems.

I. INTRODUCTION

A virtual holonomic constraint (VHC) for a mechanical system with configuration vector q is a relation of the form $h(q) = 0$ that can be made invariant via feedback. In the past decade, VHCs have emerged as a valuable tool to solve motion control problems, among them the stabilization of walking motion for biped robots pioneered by Grizzle and collaborators (see, e.g., [1]–[4]). VHCs can be used to make a haptic interface emulate the presence of an obstacle in a virtual environment or, more generally, to constrain the operator hand to move along preferred directions and to sense the virtual environment [5]. In mobile robotics, VHCs can be enforced to make a group of vehicles move in formation (see [23] for a similar notion applied to a group of kinematic vehicles). Camless combustion engines use actuated valves in place of the camshaft to regulate valve phasing, thus allowing valve phasing to be reprogrammed on-the-fly in order to optimize the operation of the engine [6]. Controlling valve phasing corresponds to replacing the mechanical constraint imposed by the camshaft with a VHC enforced by control. The use of VHCs in place of physical constraints allows greater flexibility in the design and operation of mechanical devices.

The concept of virtual holonomic constraint can be traced back to work by Paul Appell in 1911 (see [7]) and was introduced more explicitly in 1922 by Henri Beghin (see [8]) as a constraint that can be enforced through the application of external forces. The reader is referred to the work in [9], [10]. In [11], the authors investigate controlled mechanical systems for which a feedback is to enforce a holonomic constraint. They show that constrained solutions obey a differential-algebraic equation, and propose a technique to solve such equa-

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